STUDY GUIDE: DESCRIPTION OF MOTION

Kinematics of a Material Point (1D Focus)

I. Kinematics and Relativity of Motion

- **Kinematics:** A branch of mechanics that describes the motion of a body without considering the forces (causes) of the movement.
- **Relative Motion:** Motion and rest are relative. An object (e.g., a whiteboard) can be at rest relative to the classroom but in motion relative to the sun.
- **Space Frame of Reference:** An object used for comparison (e.g., the Earth).
- **Time Frame of Reference:** Used to tell the date of an event (e.g., a stopwatch or wall clock).

II. Locating an Object in 1D

1. Coordinate System & Position

For rectilinear motion, one axis $(O, \vec{i})$ is sufficient.

- **Origin of Space (O):** The point where $x = 0$ (usually the starting point).
- **Position (Abscissa):** The algebraic value $x_M = \overline{OM}$. (SI unit: meters, m).

2. Trajectory

The path followed by a moving object.

- **Rectilinear:** If the trajectory is a straight line.
- **Curvilinear:** If the trajectory is a curve.

3. Displacement vs. Distance

- **Displacement (Δx):** The change in position. $\Delta x = x_f - x_i$.
- **Note:** Δx is positive if moving right, negative if moving left, and zero if returning to start.

III. Velocity and Speed

1. Average and Instantaneous Speed

- **Average Speed (v_{av}):** $v_{av} = \frac{\text{Distance}}{\text{Time}} = \frac{d}{\Delta t}$ (SI unit: m/s).
- **Instantaneous Speed (v_i):** The speed at a specific instant t. Calculated over a very narrow time interval ($\Delta t \ll$): $v_i \approx \frac{M_{i-1}M_{i+1}}{2\tau}$.

2. Velocity Vector ($\vec{v}$)

Velocity includes both speed and direction.

Characteristics of $\vec{v}$

1. **Origin:** Current position M of the mobile.
2. **Line of Action:** The axis of motion (the rectilinear trajectory).
3. **Direction:** Same as the direction of motion.
4. **Magnitude:** The instantaneous speed v at that moment.

IV. Acceleration

1. Definitions

- **Average Acceleration (a_{av}):** Rate of change of velocity. $a_{av} = \frac{\Delta v}{\Delta t} = \frac{v_f - v_i}{t_f - t_i}$. Unit: m/s^2.
- **Instantaneous Acceleration (a_i):** $a_i \approx \frac{v_{i+1} - v_{i-1}}{2\tau}$.

2. Nature of Motion

- **Accelerated:** Velocity and Acceleration have the same direction ($v \cdot a > 0$).
- **Decelerated:** Velocity and Acceleration have opposite directions ($v \cdot a < 0$).

V. Technical Evaluation

Conversion Constant

Establishment: 1 m/s $= \frac{10^{-3}\text{ km}}{1/3600\text{ h}} = 3.6$ km/h.

Problem: A train passes the origin O at $t = 0$ and moves to point M ($x = 40$ m) in 2 s.

1. Calculate the average speed in m/s and km/h.
2. Represent the velocity vector at M using a scale of 1 cm $\rightarrow 10$ m/s.

Student Solution Space